

```
1 function E = Elastance(t,EM, Em, TC, TR)
2 %Input the following variables:
3 %t = time from 0 to T
4 %T = heart rate
5 %TC = fraction of time to go from min to max elasticity
6 %TR = fraction of time to go from max to min elasticity
7 %Em = minimum elasticity
8 %EM = maximum elasticity
9
10 % if else statement for left heart
11 if t<=TC
12     E = Em + ((EM-Em)/2)*(1-cos(pi*t/TC));
13 elseif t <=TR+TC
14     E = Em + ((EM-Em)/2)*(cos((t-TC)*pi/TR) + 1);
15 else
16     E = Em;
17 end
18
19 plot(t,E)
```